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数学建模国际赛

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我们的参赛队号为：1196

我们选择的题目是：Can Science Solve Terrorism?

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竞赛评阅编号（由竞赛评委团评阅前进行编号）：

How to Solve Terrorism with Science

Abstract

As we know, terrorism is becoming more and more serious and, thus, how to solve terrorism with science is becoming more important. This paper provides 4 mathematical models to solve the questions.

As for question one, we can build Grey Model to handle the data about net-surfing behaviors by using Matlab. After getting the personalities' information, we can build a Personality Matrix on the basis of the personalities' information. And then, we can build the relationship between Mood Matrix and Personality Matrix. In the meantime, we can build the relationship between the Mood. And then, we can acquire the equation about Emotion Matrix and Mood Matrix, Personality Matrix through Matlab. Because of emotion theory that emotion's proportions can result in different tendency to the terrorism, we can build an Emotion Model to make further processing. After acquiring data about emotion, we can use Matlab to sort, filter and handle the data in order to cancel the problem data, so we can get a approximately proper value called threshold value. This threshold value is risk index that question want.

As for the second question, we should process a huge amount of information, thus, we preprocess the network information data first, by using the fuzzy clustering algorithm. After we get the useful data, we use the Artificial Immune Algorithm (AIA) to compare the terrorists' network information data with the monitored data. Then we can get the relevance degree of everyone to the terrorists. Sorting by the relevance degree, we will have a descending list of potential terrorist name. This list can be divided into several parts, which contains terrorists' part, potential terrorists part and normal people part. And this classification is just the thing that we want.

We should carry out anti-terrorism activities from two aspects: fighting and preventing. Military fighting is inevitable and vital. Our mathematical model can apply to preventing. However, no matter how complicated the model is, there is a big problem—that is understanding terrorist behavior. Only with further psychological research can we develop a better model to describe the terrorist tendency, can we build a strategic framework to reduce terrorist behaviors. Further psychological research is a long-term investment. Except these, we should unify as a whole, increase intelligence sharing and distribute the military tasks equal to every partner's hard and soft power.

Moreover, the reliability and operability of the model have been also analyzed. Then the data analyzed is not very huge because we can attain so many precious passage, our system also can be improved.

Keywords: Grey Model, Emotion Model, Risk Index, Fuzzy Clustering Algorithm, Artificial Immune Algorithm, Psychological Research

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I. Introduction

In order to indicate the origin of terrorism, the following background is worth mentioning.

1.1 What is terrorism?

In its broadest sense, terrorism is any act designed to cause terror. In a narrower sense, terrorism can be understood to feature a political objective. The word terrorism is politically loaded and emotionally charged.

A broad array of political organizations have practiced terrorism to further their objectives. It has been practiced by both right-wing and left-wing political parties, nationalist groups, religious groups, revolutionaries, and ruling governments. The symbolism of terrorism can exploit human fear to help achieve these goals.¹

1.2 Why people turn to terrorism?

From the definition of terrorism, we know that it is used for political goal and there is no doubt that terrorists regarded as bigots possibly get hurt, even die. So why are they willing to become terrorists? This is not an easy question.

The reasons are ineluctably personal, born variously of grievance and frustration; religious piety or the desire for systemic socio-economic change; irredentist conviction or commitment to revolution.

No terrorist personality, no universal profile has been produced. But we know something about them. They are stimulated by misguided altruism or a strong sense of self-defense,

or devoted to their faith and believe that their actions are justified and divinely commanded. We have to admit that recruiters really get what the potential terrorists want.

To be specific, all terrorists consider themselves as altruists: they are doing good deeds for a wider citizens which the organization purport to represent. This sense of responsibility and self-sacrifice which helps them to justify the criminal they commit motivates people into radicalism and further terrorism. Terrorists also see them as defensive and having no alternative except taking up arms to fight against a repressive state, a predatory rival ethnic or nationalist group, or an unresponsive international order.

It seems that religion is a strong reason for justification. Yeah, it is, especially for suicide terrorism. However, it is just one more justification.

¹ Wikipedia

Individuals are attracted to terrorism in different ways. At the beginning of this war, many years ago, we can easily recognize the enemy. It was a titanic organization, located mostly in one specific geographic area, and it was led by an identifiable leader. Now, it is hard to define domestic and international terrorism; the enemy is not only identifiable organization but also individuals who may be marginalized people doing menial jobs, or some from middle and upper-middle class with good education background. Our challenge is to develop new defenses against this more amorphous, diffuse and individualized one.

II The Description of Problem

2.1 The relation between Internet usage and the tendency to be a terrorist.

We have already discussed the necessity of detecting potential individuals and terrorists. So which aspect should we get started? Today, Internet is the source of the big data, and the content of your internet usage reflects who you are. Monitoring some people's Internet use sounds like a good idea. We try to establish a risk index to assess a person's tendency.

2.2 How to process the Internet usage from monitored people?

But how to deal with the big data from so many monitored people? We preprocess the network information data first. After filtering, we get the useful data, then comparing the terrorists' network information data with the monitored data. So we can get the relevance degree of everyone to the terrorists. Sorting by the relevance degree, we will have a descending list of potential terrorist name. This list can be divided into several parts, which contains terrorists part, potential terrorists part and normal people part.

III. Models

QUESTION 1

3.1 Basic Model

3.1.1 Terms, Definitions and Symbols

In this section, we will give some basic symbols and definitions used in the following pages for convenience.

Symbols	Definitions
$n=5$	The number of evaluation objects
i	
j	Vary from 1 to 5
k	Vary from 1 to 5
$X_i=\{X_{ij} i=1,2,\dots\}$	Comparative sequence (different behavior on net-surfing)
$\sigma_i^k = \sum_{j=1}^5 f_j^k(X_{ij}) * w_j$	Comprehensive clustering coefficient about the computational object i and gray level k
$P=\{P_i i=1,2,3,4,5\}$	Different personalities
$E=\{E_i i=1,2,3,4,5\}$	Different emotions(contains anger , fear, happiness, sadness and disgust in the order)
$M=[M_P, M_A, M_D]$	Different moods(P represents pleasure; A represents arousal; D represents dominance)
T_M	Threshold value
$A=(y_1, y_2, y_3, y_4, y_5)$	A is affective variable; and y_1, y_2, y_3, y_4, y_5 represent anger, disgust, fear, happiness, sadness respectively
$w_j=[w_1, w_2, w_3, w_4, w_5],$ $w_j=\{0.2 j=1,2,3,4,5\}$	The weight coefficient of different comparative sequence
$R_i = \sum_{k=1}^n w_i * E^T$	Gray weighted correlation degree ; i represents the order of gray weighted correlation degree and t represents time is at t moment

TABLE 1

3.1.2 Assumptions

- (1) Assume $\rho=0.5$, ρ is resolution ratio which varies from 0 to 1 and the larger it is, the higher resolving power is.
- (2) Assume the weight coefficient is equal because we think the influence of the behavior on net-surfing is almost the same.
- (3) We acquire the risk index through analyzing R_i ; and we think the larger R_i is, the higher risk is.

3.1.3 The Foundation of Model

As to the task one, our group acquire the data of the people's behavior on

net-surfing ,and then adopt Gray Model to measure the probability and through the Matlab, we can assure the most possibilities of this person's personalities. In this way, we can have the passage of this people's personality and add the emotional matrix and mood matrix. Through the relationship of the emotional matrix and mood matrix and personalities matrix, we can build the Emotion Model. Through using the Matlab, we can acquire the probability that this person's tendency is to be a terrorist. And this probability is our risk index through analyzing a lot of people's data.

Procedures:

Gray Model to assure the most possibilities of this person's personalities as follows.

Step1: We should define the rules of the behavioral division.

So the rules of the behavioral division are as follows:

X1——how long you use the Internet

X2——how often you use the Internet

X3——how often you use the Internet in the region where terrorism is rampant

X4——how often you upload video to the Internet

X5——the degree of the difference in gender between in reality and virtual network

And then, we can assure the Standard sequence $X_0 = \{X_0(k) | k=1,2,3,4,5\}$.

To X1, we can divide the length about the time people on net-surfing by 24 hours

To X2, we can divide numbers that you use the Internet by 10 times

To X3, we can divide numbers that you use the Internet in the region where terrorism is rampant by 10 times

To X4, we can divide numbers that you upload video to the Internet by 10 times To X5, we can compute the quantity of the difference in gender when playing computers. And then, we divide this quantity by the total quantity in order to attain the ratio.

So, we roughly divide the behavior on net-surfing in 5 fields and we think the impact of these fields to the personalities' development is equal. So the weight coefficient $w_i = \{0.2 | k=1,2,3,4,5\}$.

Step2: We define the personality matrix $P = [p1, p2, p3, p4, p5]$.

p1 represents conscientiousness,

p2 represents agreeableness

p3 represents openness

p4 represents extraversion

p5 represents neuroticism

And then, through the OCEAN Model, we know the characteristics of this 5 elements.

Step3: We can attain A Grey Degree Based on Grey Whitening Weight Function

Assume A Grey Degree Based on Grey Whitening Weight Function as the follows:

$$f_j^1(X) = \begin{cases} 0, & X \geq 0.2 \\ \frac{X}{0.1}, & 0 < X \leq 0.1 \\ \frac{0.2-X}{0.2-0.1}, & 0.1 < X < 0.2 \end{cases} \dots(1.1)$$

$$f_j^2(X) = \begin{cases} 0, & X \geq 0.3, X \leq 0.1 \\ \frac{X-0.1}{0.2-0.1}, & 0.1 < X \leq 0.2 \\ \frac{0.3-X}{0.3-0.2}, & 0.2 < X < 0.3 \end{cases} \dots(1.2)$$

$$f_j^3(X) = \begin{cases} 0, & X \leq 0.2; X \geq 0.4 \\ \frac{X-0.2}{0.3-0.2}, & 0.2 < X \leq 0.3 \\ \frac{0.4-X}{0.4-0.3}, & 0.3 < X < 0.4 \end{cases} \dots(1.3)$$

$$f_j^4(X) = \begin{cases} 0, & X \leq 0.3; X \geq 0.5 \\ \frac{X-0.3}{0.4-0.3}, & 0.3 < X \leq 0.4 \\ \frac{0.5-X}{0.5-0.4}, & 0.4 < X < 0.5 \end{cases} \dots(1.4)$$

$$f_j^5(X) = \begin{cases} 0, & X \leq 0.4; X \geq 0.6 \\ \frac{X-0.4}{0.5-0.4}, & 0.4 < X \leq 0.5 \\ \frac{0.6-X}{0.6-0.5}, & 0.5 < X < 0.6 \end{cases} \dots(1.5)$$

diagrammatic sketch as follows:

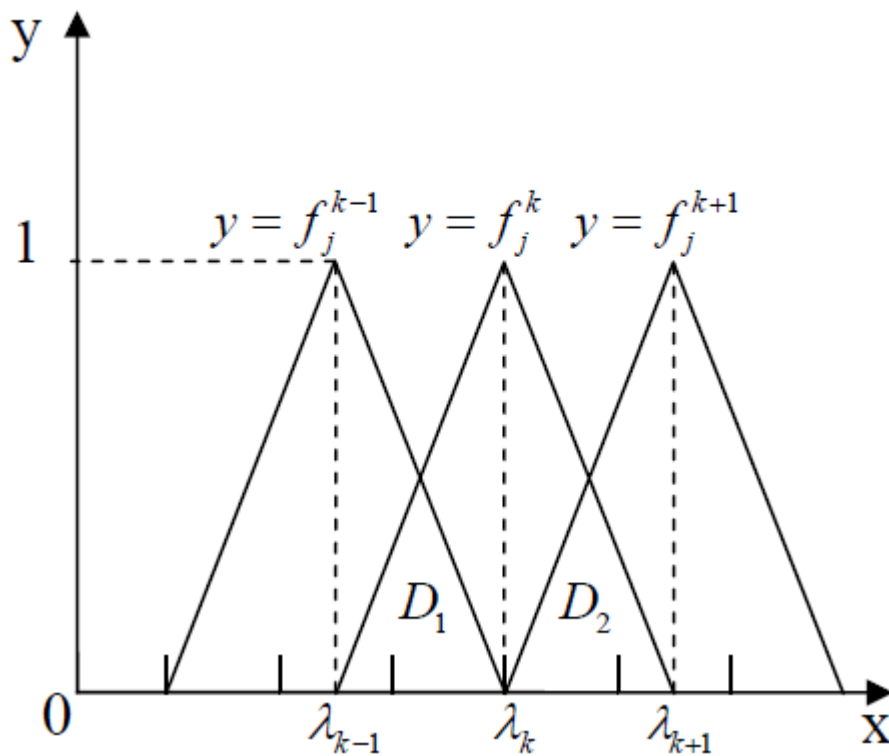


FIGURE 1

Step4: We can compute the Membership function value of gray level $f_j^k(X)$, so we

can get the σ_i^k and then, through $\max_{1 \leq k \leq 5} \{\sigma_i^k\} = \sigma_i^{k^*}$, we can attain tell the main type of the gray level.

Also, we can get the Personality Matrix $P = \{p_i = \sigma_i^k | k = 1, 2, 3, 4, 5\}$

Furthermore, this is Emotion Model to acquire the probability that this person's tendency is to be a terrorist

Step5: We can get the affective variable $A = (y_1, y_2, y_3, y_4, y_5)$.

y_1, y_2, y_3, y_4, y_5 represent anger, disgust, fear, happiness, sadness respectively

And we define its mathematical expression as follows:

$$y = \begin{cases} a_1 * t^{b_1} & 0 \leq t \leq t_1 \\ a_2 * e^{-b_2(t-t_2)} & \dots \end{cases} \quad (1.6)$$

Also we can build emotional matrix $E = \{e_i(k) | k = 1, 2, 3, 4, 5\}$

mood matrix $M = [M_P, M_A, M_D]$

personality matrix $P = \{p_i(k) | k = 1, 2, 3, 4, 5\}$

Step6: We can acquire the transformation between the mood and personality.

The mathematical expression is following:

Pleasure = $0.21 * \text{extraversion} + 0.59 * \text{agreeableness} + 0.19 * \text{neuroticism}$

Arousal = $0.15 * \text{openness} + 0.30 * \text{agreeableness} - 0.57 * \text{neuroticism}$

Dominance = $0.25 * \text{openness} + 0.17 * \text{conscientiousness} + 0.60 * \text{extraversion} - 0.32 * \text{agreeableness}$

So, obviously, $M = K * P$

Inside $K = [0, 0, 0.21, 0.59, 0.19 ; 0.15, 0, 0, 0.30, -0.57 ; 0.25, 0.17, 0.6, 0.32, 0]$

Step7: We can acquire the transformation between emotion and mood, personality.

The diagram about transformation between motion and mood as follows:

emotion	Pleasure	Arousal	Dominance	Mood subspace
Anger	-0.51	0.59	0.25	-P+A+D
Disgust	-0.40	0.20	0.10	-P+A+D
Fear	-0.64	0.60	-0.43	-P+A-D
Happiness	0.40	0.20	0.15	+P+A+D
Sadness	-0.40	-0.20	-0.50	-P-A-D

TABLE 2

But mood change intensity is a continuous quantity varying from 0 to 1.

So, we can define the mapping base in the mood space of the motion:

$$\begin{aligned}
 PAD^* &= [PAD_{Anger}, PAD_{Disgust}, PAD_{Fear}, PAD_{Happiness}, PAD_{Sadness}] \\
 &= \begin{bmatrix} -0.51 & -0.40 & -0.64 & 0.40 & -0.40 \\ 0.59 & 0.20 & 0.60 & 0.20 & -0.20 \\ 0.25 & 0.10 & -0.43 & 0.15 & -0.50 \end{bmatrix}
 \end{aligned}$$

...(1.7)

So, we can get the certain transformation:

$$E=f(M,PAD^*)=D/\sum_{i=1}^5 d_i \dots(1.8)$$

Inside, $D=[d_1, d_2, d_3, d_4, d_5]$; $d_i = [(M - PAD_i)^T(M - PAD_i)]^{\frac{1}{2}}$, $i=1,2,3,4,5$ and correspond to anger, disgust, fear, happiness, sadness

Step7: We can adopt E and w_j in order to acquire the R_i through $R_i=\sum_{k=1}^n w_i * E^T$.

By analyzing a lot of data from different people, we can get a lot of different values of R_i . And then, we can get a threshold value T_M which we can also call it risk index.

3.1.4 Analysis of the Result

As we know, we can get various data by monitoring people using the Internet. And then, we can attain many result which can imply the final result about tendency to the terrorism. However, different people have different value of R_i . So, we need adopt predict model to deal with the relationship between R_i and T_M .

First: We can sort data and then discover common grounds of them. But the quantity of data is so large and obviously there are many improper data, we need filter the data.

Second: After sorting the data, we need to filter the improper data. And then, we get some meaningful data. Assume the quantity of data we get is m . So, we can get the T_M .

$$T_M=\sum R_i./m\dots(1.9)$$

Third: After getting the value of T_M , we can give the approximate value of the risk index. whose index is higher than the risk index, we can identify a person whose index is higher than the risk index as a potential terrorist.

3.1.5 Strength and Weakness

Strength:

- (1) We adopt Gray Model and Motion Model to get the Gray weighted correlation degree R_i and consider many factors such as net-surfing behaviors, mood, personality and emotion. These can make the result more precious.
- (2) We use Gray weighted correlation degree R_i to get threshold value. And with a lot of data, this method that attain the risk index can be more convincing.
- (3) In the procedure of analyzing data, we filter some problem data. As we know, some people can get the wrong passage, so the data we get may have some problems. If we do not abandon them, these may have a negative impact on our final result.

Weakness:

- (1) Because we cannot acquire the exact data people surfing the Internet, we cannot get the exact final risk index.

(2) Because of lacking exact data, the weight we set up may be not very precious.

QUESTION 2

3.2 Basic Model

3.2.1 Terms, Definitions and Symbols

Symbols	Definitions
$Z = \{ Z_1, Z_2, Z_3, \dots Z_q \}$	network information data(NID) set
$B = \{ b_1, b_2, b_3, \dots b_q \}$	NID's distribution center set
f_{kl}^2	NID's Proportion
y_{kl}	NID's update
e, q, p, d	y_{kl} 's state parameter
b_l	Potentialterrorist's average NID
P_n	probability set of each person to be a terrorist
P_d	P_n after sorted
I'	confidence level
ξ_{lm}	proportion of the potentialterrorists
Z_{lm}	relevance degree of the Potentialterrorists
ε	the threshold of the potentialterrorists relevance
D_j	antigen set
C_d	antibody set

TABLE 3

3.2.2Assumptions

1. Assume that the government can monitor the Internet server that we are interested in.
2. Assume that the network information data is about the login time, last time, login frequency and data size.

3.2.3The Foundation of Model

Different with the basic model, we do not use the “Emotion Model(情感模型)” to process the Network information data.And we put forward the Artificial Immune Algorithm (AIA) to analyze the Network information data.

The AIA goes like this:

- **The first step: Data preprocessing**

Because the network information data is really huge, so we try to classify the data and

cluster some similar together roughly. And then calculate the distance of network information data to the clustering center.

- **The second step: Set the Antigen and Antibody of the network information data**

The antigen is the information about the terrorism that we are interested in, and the antibody is the specificity algorithm that distinguishes the antigen from the whole network information data. In this step we determine the antigen and the antibody of our model.

- **The third step: Compute the matching degree of the antigen and the antibody**

In this step, we compute the matching degree of the interested information(antigen) and thenetwork information data(antibody).

- **The last step: Sort the result of the third step and get the conclusion**

After getting the result of the matching degree, we sort the result from big to small. And then we can classify the result into different classes according to the sorted result.

This is the basic method of the AIA, however, how can we preprocess the network information data and compute the matching degree? We consider the data preprocessing in the following way:

The way to preprocess the data:Fuzzy clustering processing

- Using the set $Z = \{ Z_1, Z_2, Z_3, \dots, Z_q \}$ to represent the network information data, and we can use the following formula to describe the object function of the network information data:

$$L_P(Y, B) = \sum_{k=1}^q \sum_{l=1}^e y_l^p f_{kl}^2(z_k, b_l) \dots (2.1)$$

In this formula, $B = \{ b_1, b_2, b_3, \dots, b_q \}$, b_k is used to describe the distribution center of the k th group of the network information data, f_{kl}^2 is use to describe the different kinds of network information data's Proportion . According to the following formula , we can calculate the distance from information to distribution center:

$$f_{kl}(z_k - b_l) = |z_f - b_f| \dots (2.2)$$

- Set “ e, q, p, d” as the state parameter , and “d” is initialized to 1. The distribution center of the whole network information set is $B(d) = (b_1, b_2, \dots, b_e)$, according to the following formula we can updatethe network information data:

$$y_{kl} = \frac{1}{\sum_{m=1}^e \left[\frac{f_{kl}}{f_{km}} \right]^{\frac{2}{p-1}}} \text{when } f_{kl} \neq 1 \dots (2.3)$$

$$y_{kl} = 0 \quad \text{when } e_{k1} = 0, l \neq m$$

$$y_{kl} = 1 \quad \text{when } e_{k1} = 0$$

- According to the following formula, we can compute the potentialterrorist's

average network information data

$$b_l = \frac{\sum_{k=1}^q y_{kl}}{\sum_{k=1}^q y_{lm}} \dots (2.4)$$

Comparing this result with the next average network information data, if it satisfy the following condition , then complete the data preprocess , if not, we have to repeat the above step until reach the maximum time of the process.

$$|b_d - b_{d+1}| \leq \varphi \dots (2.5)$$

Using the above method we can preprocess the data.

The way to compute Matching degree: Matching degree computing algorithm

- Assume the probability set of each person to be a terrorist as P_n , if the network information data $|P_n| < n-1$, we analyze the remaining $n-1$ groups of network information data. And the $|P_n|$ is the most likely group of people that can be a terrorist. Supposing the amount of information of $|P_n|$ is less than $n-1$, therefore, this group of data do not include a terrorist.
- Then we need to sort the probabilityset, and we use $P_d = \{P_c[1], P_c[1], P_c[3], \dots, P_c[n+1]\}$ to represent for the result.
- According to the association rules of the network information analyzing, we can get the following data that relates to terrorists:

$$F1 = \{(C,2),(D,4), \dots, (G,6)\} \dots (2.6)$$

From the huge amount of people, we choose the kind of people whose characteristic value is larger than the threshold, and we use $P1 = \{C, D, E, F, G\}$ to represent it. Then analyzing the set of this people again, we can get the set of people whose probability to be a terrorist is biggest, using $P2 = \{CD, CE, CF, DE, DF, DG, EF, EG\}$ to represent them.

- Then we can use the following formula to compute the confidence level of the Potentialterrorists:

$$I' = \frac{\sum_{l=1}^S [\prod \xi(g_l, a_m)]}{r} \dots (2.7)$$

Then use the following formula to compute the proportion of the potentialterrorists in the society:

$$\xi_{lm} = \frac{g(f_l, a_m)}{\sum_{m=1}^p g(f_l, a_l)} \dots (2.8)$$

Next use the following formula to compute the relevance degree of the Potentialterrorists:

$$Z_{lm} = \frac{\exp(\frac{-x}{\lambda})}{\sum_{l=1}^q \exp(\frac{-x}{\lambda})} \dots (2.9)$$

- Set the threshold of the potential terrorists relevance to ε . Then we can use the following formula to find the potential terrorists :

$$\left. \begin{array}{l} f_l > \varepsilon \quad \text{this person is a terrorist} \\ f_l < \varepsilon \quad \text{this person is not a terrorist} \end{array} \right\} \dots (2.10)$$

Using the above algorithms we can compute the relevance degree.

3.2.4 Solution and Result

In the Foundation of the model, we have talked about the abstract method of how to distinguish the terrorist, but we have not implemented the concrete method. So in this part, we will show you the concrete method.

When we implement the AIA, the first step, we describe the antigen set as following:

$$D_j = (D_{j1}, D_{j2}, \dots, D_{j3}), (l = 1, 2, \dots, q) \dots (2.11)$$

The D_j represents for the i th internet user's network information data. A user may have several kinds of network information data, thus, the D_j is a class that contain several network information data.

$$D_{ji} = (a_1, a_2, \dots, a_i) \dots (2.12)$$

Assume that if $a_i=1$ means the user always visits one internet server, such as: YouTube, Msn etc., and if $a_i=0$ means that the user seldom visits one internet server. The government can monitor the internet server, for example the USA have the "Prism plan"(棱镜计划), so it is easy for government to get the necessary information.

The second step: we describe the antibody set as following:

$$C_d = (C_{d1}, C_{d2}, \dots, C_k), (k = 1, 2, \dots, a) \dots (2.13)$$

The C_{di} represent for the standard network information data of terrorists, different kind of terrorists may have different kind standard network information data, thus, we prepare for different kind of standard network information data for different kind of terrorists which means that the number a is not fixed, we can change it as we need.

The third step: after we get the above information data, we can use the Matching Degree Computing Algorithm to classify different kind of people.

3.2.5 Strength and Weakness

Strength:

- (1) In this model, we analyze the habit of the terrorist using internet, and we set this habit as the antibody. Thus, this model can imitate the specific immunity of the human body, and it can distinguish the terrorist more accurately.
- (2) What's more, because the network information data is a huge amount of information, and it may come from different sources, therefore, before we process the data, we should preprocess the data to make data amount smaller and unify the format of the data, this model uses the fuzzy clustering algorithm to preprocess the data, and do a perfect job.

Weakness:

- (1) If we want to analyze the habit of the terrorists' habit, we need to monitor the terrorist for a long time, which may be not allowed in the reality. And the key of the model is to set the appropriate antibody, thus, if we don't choose the most suitable antibody, the model may be no longer useful.
- (2) Moreover, if we use this model, we have to monitor a huge amount of data, which means that we have to process the same amount of data, it is a problem, what if the computer cannot process the data timely?

QUESTION 3

3.3 Advice on fighting terrorism

The counter-terrorism work needs long-term investment. We should carry out from two aspects: fighting and preventing. Military fighting is inevitable and vital. In terms of the situation nowadays, if they never pay a heavy price and their core members are not destroyed, the organization will exit. How to fight back in the front-line battlefield to deter them is not the problem that we can give suggestions. But we can exploit the mathematical models to detect people who may convert into a terrorist. However, there is still a big problem –that is understanding terrorist behavior. Only with further psychological research can we develop a better model to describe the terrorist tendency, can we build a strategic framework to reduce terrorist behaviors. Further psychological research is a long-term investment.

Also, fighting terrorism needs all countries' efforts. How to unify all the power, increase intelligence sharing and distribute the military tasks equal to every partner's hard and soft power involves game theory and international negotiation.

It is a complicated problem in macroscopic view. Recognizing target is another important issue: to strike the terrorists only, Daesh and the jihadi groups only, and not to strike the forces and the groups that are fighting against the terrorists. To maintain and cultivate local armed forces, it is crucial to provide weapons and training for them fighting against ISIS in Iraq and al Shams.

How to tackle with terrorism without having negative effect on a country and its people is worth considering in microscopic view.

IV Conclusions

4.1 Conclusions of the problem

For the question one, we used the grey model and emotion model to get the risk index, which is used to judge whether a specific person is terrorist or not. This basic model considers many factors such as net-surfing behavior, mood, personality and emotion. All this effort makes the risk index more precious, nevertheless, the lack of data makes us in a dilemma, we have to do more research about it.

As to the second question, we use the fuzzy clustering algorithm to preprocess the huge network information data, use the artificial immune algorithm to get the relevance degree of everyone to the terrorists. Both the fuzzy clustering algorithm and the artificial immune algorithm make it more effective and accurate to classify the Internet user. The key of the model is antibody that we choose, if we choose a useless antibody, it may increase the data that we need to process and get a bad result.

4.2 Methods used in our models

In our model we have used the grey model, emotion model, fuzzy clustering algorithm, artificial immune algorithm. The concrete introduction of the methods is as following:

4.2.1 Grey model

We think that different personalities can result in different net-surfing behaviors. So we can build Grey Model to handle the data about net-surfing behaviors by using Matlab. We need input behavior data into Grey Model, and we can get the personalities' information

4.2.2 Emotion model

According to the personalities' information we can build a Personality Matrix. We can build a Mood Matrix and Emotion Matrix according to the psychological theory. On the top of that, we can build the relationship between Mood Matrix and Personality Matrix. On the top of that, we can build the relationship between the Mood. And then, we can acquire the equation about Emotion Matrix and Mood Matrix, Personality Matrix through Matlab. And according to emotion theory, we divide emotions of one person into fear, anger, disgust, happiness and sadness. And we consider different emotion proportions can result in different tendency to the terrorism. Because of that, we can build an Emotion Model to make further processing in order to attain the risk index. So, when we input the personalities' information, we can acquire the risk index.

4.2.3 Fuzzy clustering algorithm

It is a data preprocess algorithm, which can make the different kinds of data that comes from different ways into the same format. And this algorithm ignores some unimportant data and put some data together, all this effort makes the data easier to process.

4.2.4 Artificial immune algorithm

The artificial immune algorithm is used to compare the terrorists' network information data with normal people. After the data preprocessing, we can set the data of the terrorists as the object data (antibody), and the algorithm can imitate the specific immunity of the human body, which means that it can automatic distinguishing.

V. Future Work

5.1 Advantages of the model

- (1) Our system clearly make use of the data and using Matlab respectively to deal with the data of persons' behavioral data.
- (2) This paper uses the Emotion Model and Grey Model and Artificial Immune Algorithm to analyze the Network information. According to this system, we can clearly solve the complex problem.
- (3) In this paper, using chart to illustrate the problems in the proper place, thus we can explain the problem more clearly.

5.2 Disadvantages of the modeled

- (1) As to weight coefficient, we need a lot of data as a base. But, clearly, we cannot acquire so many data, so the weight coefficient we set up in the paper cannot be very precious. It may have a little influence on the result we get through the models.
- (2) According to the title, we only consider Network information, but in fact, there are many factors have an impact on the somebody's tendency to the terrorism such as atmosphere, education and relationship with others. Although Network information can reflect some aspects, they cannot reflect the whole things.
- (3) Although we use Matlab to deal with the data we get, the velocity of handling data is not very high and we need some time for waiting for the result.

5.3 Improved method of the model

- (1) We can increase the approaches getting data not only monitoring people's network information. Because there are many factors have an impact on the somebody's tendency to the terrorism such as atmosphere, education and relationship with others. And these factors all affect the somebody's tendency to the terrorism and these information have value for us to acquire.
- (2) We can use more advanced appliance to deal with our data in order to increase the velocity of the handling and it is beneficial for us to detect early and treat in time. And through exploiting more approaches to get information, we have much more data to do research. So it is necessary to improve equipment performance.
- (3) We can add some examining models to test our results in order to make other results more precious which can get rid of leaving out some terrorist hiding in the common people and misdeeming common people as terrorists.

VI. References

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VII. Appendix

Code of GreyModel

```
% Greymodel.m

function P=Greymodel()

P=zeros(1,5);

global k;

for k=1:5

totalcluster=0;

for j=1:5

zhibiao=X(j);

Greywhiten=white(zhibiao);

cluster=Greywhiten*0.2;

totalcluster=totalcluster+cluster;

end

P(k)=totalcluster;

end

% white.m

function Greywhiten=white()

global k;

switch(k)

case 1

if(zhibiao>=0.2)

Greywhiten=0;
```

end

if($0 < \text{zhibiao} \&\& \text{zhibiao} < 0.1$)

Greywhiten= $\text{zhibiao}/0.1$;

end

if($0.1 < \text{zhibiao} \&\& \text{zhibiao} < 0.2$)

Greywhiten= $(0.2 - \text{zhibiao})/(0.2 - 0.1)$;

end

case 2

if($\text{zhibiao} \leq 0.1 \parallel \text{zhibiao} \geq 0.3$)

Greywhiten=0;

end

if($0.1 < \text{zhibiao} \&\& \text{zhibiao} < 0.2$)

Greywhiten= $(\text{zhibiao} - 0.1)/(0.2 - 0.1)$;

end

if($0.2 < \text{zhibiao} \&\& \text{zhibiao} < 0.3$)

Greywhiten= $(0.3 - \text{zhibiao})/(0.3 - 0.2)$;

end

case 3

if($\text{zhibiao} \leq 0.2 \parallel \text{zhibiao} \geq 0.4$)

Greywhiten=0;

end

if($0.2 < \text{zhibiao} \&\& \text{zhibiao} < 0.3$)

Greywhiten= $(\text{zhibiao} - 0.2)/(0.3 - 0.2)$;

end

```
if(0.3<zhibiao&&zhibiao<0.4)

Greywhiten=(0.4-zhibiao)/(0.4-0.3);

end

case 4

if(zhibiao<=0.3||zhibiao>=0.5)

Greywhiten=0;

end

if(0.3<zhibiao&&zhibiao<0.4)

Greywhiten=(zhibiao-0.3)/(0.4-0.3);

end

if(0.4<zhibiao&&zhibiao<0.5)

Greywhiten=(0.5-zhibiao)/(0.5-0.4);

end

case 5

if(zhibiao<=0.4||zhibiao>=0.6)

Greywhiten=0;

end

if(0.4<zhibiao&&zhibiao<0.5)

Greywhiten=(zhibiao-0.4)/(0.5-0.4);

end

if(0.5<zhibiao&&zhibiao<0.6)

Greywhiten=(0.6-zhibiao)/(0.6-0.5);

end

end
```